CYS-262L — INFORMATION SECURITY LAB

Lab 2: Data & Asset Classification Exercise

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Lab / Assignment	2 — Data & Asset Classification

Part B — Classification Schema

The four-tier schema below classifies records according to the consequence and sensitivity of disclosure, while also considering privacy, legal, financial, safety, integrity, and availability impacts.

Tier	Definition / Criteria	Who May Access	Handling Requirement
Tier 1 — Public	Information approved for unrestricted external release; disclosure causes no meaningful harm.	Anyone may access it.	May be published on the public website; encryption is not required for ordinary public distribution.
Tier 2 — Internal	Information intended for use within the institution; disclosure may cause minor inconvenience but not serious harm.	Faculty, staff, and authorized institutional users.	Share through institutional accounts/systems and do not post publicly.
Tier 3 — Confidential	Information whose disclosure could cause meaningful harm to individuals or the institution.	Only named roles or authorized personnel with a legitimate need to know.	Encrypt in transit and at rest; store only in approved institutional systems.
Tier 4 — Restricted	Highly sensitive information whose disclosure could cause severe legal, financial, safety, or privacy harm.	Named individuals only, strictly on a need-to-know basis.	Encryption is mandatory, access must be logged, and secure disposal is required.

Part C — Classification of the 25 Records

Record #	Record Description	Assigned Tier	Justification (where applicable)
1	Public course catalog listing for the current semester	Tier 1 — Public	
2	An individual student's final transcript	Tier 3 — Confidential	Contains an identifiable student's academic record and should be protected from unauthorized disclosure.
3	Draft copy of next week's midterm exam paper	Tier 3 — Confidential	Premature disclosure could compromise assessment integrity and fairness.
4	University's official press release announcing a new campus	Tier 1 — Public	
5	Payroll record listing a faculty member's salary	Tier 3 — Confidential	Contains personal financial/employment information that could cause privacy harm if disclosed.

6	Campus building floor plan showing server room location	Tier 3 — Confidential	Detailed security-sensitive information could facilitate physical targeting of critical infrastructure.
7	Anonymized aggregate survey results on student satisfaction	Tier 1 — Public	
8	A student's disability/medical accommodation request	Tier 4 — Restricted	Highly sensitive personal/medical information with severe privacy implications.
9	Faculty meeting minutes discussing a routine scheduling matter	Tier 2 — Internal	Routine institutional information is appropriate for internal use but not necessarily public release.
10	Login credentials for the university's admissions portal	Tier 4 — Restricted	Credentials provide direct access to a protected system and require strict access control.
11	Alumni donor contact list with donation amounts	Tier 3 — Confidential	Contains personal contact and financial/donation information.
12	Publicly available academic calendar (holidays, deadlines)	Tier 1 — Public	
13	Incident report describing a network intrusion attempt	Tier 3 — Confidential	Security incident details could expose defensive information or create institutional risk if broadly disclosed.
14	A professor's unpublished research dataset (pre-publication)	Tier 3 — Confidential	Unpublished research may have intellectual-property and competitive implications.
15	Signed vendor contract with a cloud hosting provider	Tier 3 — Confidential	Contractual and commercial information may create institutional or financial harm if improperly disclosed.
16	Security camera footage from a campus parking lot	Tier 3 — Confidential	Footage may identify individuals and reveal security-related activity or patterns.
17	Student club's public event flyer	Tier 1 — Public	
18	List of Wi-Fi passwords for staff-only network segments	Tier 4 — Restricted	Disclosure would enable unauthorized access to protected network resources.
19	A rejected job applicant's resume and interview notes	Tier 3 — Confidential	Contains personal and employment-related information about an identifiable applicant.
20	University's annual public financial summary report	Tier 1 — Public	
21	Internal IT ticket describing a minor printer malfunction	Tier 2 — Internal	Routine operational information intended for

			institutional use and not public posting.
22	Departmental budget draft not yet approved by finance	Tier 3 — Confidential	Unapproved financial planning information could cause institutional or decision-making harm if prematurely disclosed.
23	A student's disciplinary/conduct case file	Tier 4 — Restricted	Contains highly sensitive identifiable student information and potentially serious privacy consequences.
24	Class syllabus for CYS-262 posted on the LMS	Tier 2 — Internal	The manual identifies it as posted on the LMS; absent a public-release designation, it is treated as institutional course information.
25	Database backup file containing all student records	Tier 4 — Restricted	A complete student-record database is highly sensitive and requires strict, logged, encrypted access.

Tier Summary

Tier	Number of Records
Tier 1 — Public	6
Tier 2 — Internal	3
Tier 3 — Confidential	11
Tier 4 — Restricted	5

Part D — Ambiguous / Borderline Cases

Record	Justification
Record 6 — Server-room floor plan	Classified as Confidential because the manual says classification should consider security impact. A floor plan that identifies a server room is not necessarily secret, but wider disclosure could increase physical-security risk.
Record 14 — Unpublished research dataset	Classified as Confidential because it is pre-publication and may have intellectual-property or competitive value. If the dataset were fully anonymized and explicitly intended for open release, a lower tier could be justified.
Record 22 — Departmental budget draft	Classified as Confidential because it is an unapproved financial planning document. Its sensitivity may change after approval and public release.
Record 24 — CYS-262 syllabus on the LMS	Classified as Internal because the source only states that it is posted on the LMS, not that it is approved for unrestricted public release. If the institution publicly publishes the syllabus, Tier 1 would be appropriate.